

	$\frac{1.2857}{A_y} = 1.2857 \approx 1.3$
5(a)	The resistance of the thermistor is the ratio of the potential difference (p.d.) across it to the current flowing through it. As the p.d. is increased from zero, the graph is a straight line passing through the origin. Hence the resistance of the thermistor is constant, As the p.d. increases, after a certain point, the graph starts to curve upwards. The ratio of the p.d. to current decreases. Hence the resistance of the thermistor decreases.
5(b)(i)	Since the resistors are connected in series, the current I flowing through both of them is the same. $\frac{\text{power dissipated in } 220 \, \Omega \text{ resistor}}{\text{power dissipated in } 640 \, \Omega \text{ resistor}} = \frac{I^2 R_1}{I^2 R_2}$ $= \frac{220}{640}$ $= 0.34375$ $\approx 0.34 \, \Omega \text{ (2 s.f.) (shown)}$
5(b)(ii)	The ratio of the powers in (b)(i) remains the same at 0.34. This is because the two resistors are still connected in series and the current through both of them

	will be the same, even though the current is lower than when the supply has negligible internal resistance. The current is cancelled in the ratio of powers, meaning that the current does not influence the ratio.
5(b)(iii)	Since $I = \frac{V}{R}$, when the potential difference is constant, $I \propto \frac{1}{R}$ $\frac{\text{current from supply in series circuit}}{\text{current from supply in parallel circuit}} = \frac{\text{effective resistance of parallel circuit}}{\text{effective resistance of series circuit}}$ $= \frac{\left(\frac{1}{220} + \frac{1}{640} \right)^{-1}}{220 + 640}$ $= 0.19 \text{ (2 s.f.)}$
5(b)(iv)	The ratio of the currents in (b)(iii) remains the same at 0.19. As the same a.c. power supply is used for both circuits, the root-mean-square voltage of the power supply will be the same for both circuits. As such the relationship $I \propto \frac{1}{R}$ still holds true and the calculation in (b)(iii) and hence the ratio is the same.
6(a)	Faraday's law of electromagnetic induction states that the induced e.m.f. in a